

Detailed Errata and Statement Audit

Three chemotaxis–logistic papers formalized in Lean 4

ShenWork formalization project

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Abstract

This document records statement defects and proof gaps found while formalizing three papers on parabolic–elliptic chemotaxis systems. It deliberately separates four logically different outcomes: an explicit counterexample to a printed conclusion, a gap in the printed proof without a counterexample to the conclusion, a wording clarification, and a defect in an earlier Lean interface that is not an error in the paper. For every mathematical counterexample we give the calculation, the corrected statement, and the exact Lean certificate. The repository is pinned to Lean 4.30.0 and Mathlib v4.30.0. The cited certificates are `sorry`-free and their headline axiom audits contain only `propext`, `Classical.choice`, and `Quot.sound`.

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1 Scope, sources, and classification

The audited sources are:

1. Wenxian Shen, *Existence, uniqueness, stability, and monotonicity of traveling waves for repulsion/attraction chemotaxis models with logistic type source*, arXiv:2605.04401v1 (the “traveling-wave paper”);
2. Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence*, arXiv:2512.14858v1 (“Part I”);
3. Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, II: Persistence and stabilization*, arXiv:2604.02599v1 (“Part II”).

The labels used below have precise meanings.

False statement

The printed hypotheses admit an explicit object for which the printed conclusion fails.

Proof gap

A displayed step is invalid or a required hypothesis is missing from the proof. No counterexample to the theorem itself is asserted.

Clarification

The mathematical intent is recoverable, but a literal reading is stronger or less precise than the continuation or regularity framework supports.

Interface only

An earlier Lean definition quantified over more objects, or stored less semantic information, than the paper. A refutation of that Lean definition is not evidence against the paper.

#	Source	Finding	Status
E1	Traveling wave	Theorems 1.2–1.3: seven defects in the Section 5 weighted-energy argument	PROOF GAP; corrected proof window
E2	Traveling wave	Lemma 4.2: the constant subsolution is not valid for every member of the printed trap set	FALSE STATEMENT
E3	Part I	Theorem 1.2 admits $a > 0, b = 0$, where constant solutions grow exponentially	FALSE STATEMENT
E4	Part I	Theorem 1.3(iv) invokes an elliptic estimate outside its exponent domain	PROOF GAP
E5	Part I	Proposition 1.1: the endpoint alternative belongs to the maximal continuation, not an arbitrary finite witness	CLARIFICATION
E6	Part I	Lemma 2.7 controls only $t > 0$ but concludes a bound including $t = 0$	FALSE STATEMENT as written
E7	Part II	Theorem 2.1(1) admits the pure-decay regime $a = 0 < b$	FALSE STATEMENT

#	Source	Finding	Status
E8	Part II	Theorems 2.2–2.5 reuse an all-time C^1 estimate from merely continuous or L^∞ -small data	CLARIFICATION; ill-posed at $t = 0$

2 Traveling-wave paper

2.1 E1: Theorems 1.2–1.3, weighted stability

Printed claim. Let

$$\kappa = \frac{c - \sqrt{c^2 - 4}}{2}, \quad \kappa^2 - c\kappa + 1 = 0.$$

Theorem 1.2 states weighted stability for

$$\kappa < \eta < \frac{1}{1 + |\chi|^{1/6}},$$

and (1.21) is printed with the laboratory-coordinate weight $e^{2\eta x}$. Theorem 1.3 uses the same stability theorem in its uniqueness argument.

The load-bearing root error. The coefficient produced by (5.31) is

$$q(\eta) = \eta^2 - (c - A)\eta + (1 + B),$$

where the displayed quantities in (5.32)–(5.33) have $A \geq 0$ and, for $\chi \neq 0$, $B > 0$. Evaluation at the unperturbed tail exponent is exact:

$$q(\kappa) = (\kappa^2 - c\kappa + 1) + A\kappa + B = A\kappa + B > 0.$$

Therefore κ is not between the two roots of q . Under the speed condition it lies below the lower perturbed root

$$\kappa^- = \frac{(c - A) - \sqrt{(c - A)^2 - 4(1 + B)}}{2}.$$

Continuity of q then gives $\eta > \kappa$, arbitrarily close to κ , for which $q(\eta) > 0$. The global coefficient estimate written in Section 5 proves decay only for $\eta > \kappa^-$, not for every $\eta > \kappa$. This is a proof gap, not a constructed unstable solution: a new localized-coercivity or weighted spectral argument could conceivably recover the larger interval.

Coordinate and exponential signs. The energy actually estimated after changing to $z = x - ct$ is

$$E_{\text{move}}(t) = \int_{\mathbb{R}} e^{2\eta z} |u(t, z + ct) - U(z)|^2 dz.$$

The printed (1.21) is

$$E_{\text{lab}}(t) = \int_{\mathbb{R}} e^{2\eta x} |u(t, x) - U(x - ct)|^2 dx.$$

Changing variables $x = z + ct$ yields

$$E_{\text{lab}}(t) = e^{2\eta ct} E_{\text{move}}(t).$$

Thus decay of the moving-frame quantity does not by itself imply the printed laboratory-weighted limit. The correct laboratory notation is $e^{2\eta(x-ct)}$. In addition, the paper declares $\lambda = q(\eta) < 0$ and then uses $e^{-\lambda t}$ as a decaying factor. For this sign convention the decaying factor is $e^{\lambda t}$, or, equivalently, $e^{-\rho t}$ with $\rho = -\lambda > 0$.

Four coefficient errors. Writing $w = u - U$, $z = v - V$, the elliptic relation $v_{xx} = v - u^\gamma$ gives the zero-order chemotactic flux difference

$$(va_m - a_{m+\gamma})w + U^m z.$$

After multiplication by $-\chi$, the perturbation equation contains

$$-\chi(va_m - a_{m+\gamma} + b_2)w - \chi b_4 z.$$

Equations (5.18)–(5.19) reverse both the $a_{m+\gamma}$ and $b_4 z$ signs. These signs can be repaired by absolute-value estimates. Three further estimates must also be corrected:

$$\begin{aligned} |\chi b_1 w_x w| &\leq \frac{1}{2} w_x^2 + \frac{1}{2} |\chi|^2 b_1^2 w^2, & \text{not a term linear in } b_1, \\ \int_{\mathbb{R}} V^2 &\leq \frac{\gamma^2 M^{2(\gamma-1)}}{(1-\eta)^2} \int_{\mathbb{R}} W^2, \\ \int_{\mathbb{R}} V_x^2 &\leq \frac{\gamma^2 M^{2(\gamma-1)}}{1-\eta^2} \int_{\mathbb{R}} W^2. \end{aligned}$$

The factor $M^{2(\gamma-1)}$ is absent from (5.29)–(5.30). Finally, Lemma 5.2 proves only $U_x/U \leq C$, while (5.23) uses $|U_x/U| \leq C$. The latter requires a two-sided logarithmic-derivative estimate.

Corrected form and Lean evidence. The formalization retains the full corrected budget A_χ, B_χ , the root window

$$\kappa_{c,\chi,m,\alpha,\gamma}^- < \eta < \frac{1}{1 + |\chi|^{1/6}},$$

and moving-frame weighted convergence. The main certificates are:

- `paper531_kappa_not_between_perturbed_roots`, `paper531_kappa_lt_rootMinus`, and `paper531_positive_inside_stated_weight_window` in `Paper1/Theorem12RootObstruction.lean`;
- `laboratoryWeightedL2Energy_eq_exp_mul_coMoving` in `Paper1/Theorem12CoordinateAudit.lean`;
- the corrected flux identity in `Paper1/Theorem12MeanCoefficients.lean`;
- the squared Young term and resolver cap in `Paper1/Theorem12WeightedEnergy.lean`;
- the two-sided Riccati fencing estimate in `Paper1/Theorem12LogDerivative.lean`.

The corrected headline theorems are `paper1_Theorem_1_2_chi_pos_unconditional`, `paper1_Theorem_1_2_chi_zero_unconditional`, and `paper1_Theorem_1_2_chi_neg_unconditional`. Their quantified weight range begins at the corrected `paper531RootMinus`.

2.2 E2: Lemma 4.2, constant subsolution

Printed claim. For $0 < \kappa < \tilde{\kappa}$ in the displayed parameter range, $M \geq 1$, $D > D_{M,\kappa,\tilde{\kappa},\chi,m,\gamma}$, every $u \in E_{\kappa,M,T}$, and every

$$0 < d \leq d_{\kappa,\tilde{\kappa},D,\chi},$$

Lemma 4.2(2) asserts that the constant function $W = d$ is a subsolution on the whole line.

Counterexample. Suppress time and take the admissible trap profile

$$u(y) = \min\{1, e^{-y/2}\} \min\{1, |y + 1|\}.$$

It is continuous, nonnegative, bounded by the printed upper barrier for $\kappa = \frac{1}{2}$, $M = 1$, and satisfies

$$u(-1) = 0, \quad u(0) = 1.$$

For $m = \alpha = \gamma = 1$, the frozen elliptic response at -1 is

$$V_0 = \frac{1}{2} \int_{\mathbb{R}} e^{-| -1 - y |} u(y) dy > 0.$$

Choose

$$\chi = \frac{2}{V_0}, \quad \kappa = \frac{1}{2}, \quad \tilde{\kappa} = 1, \quad M = 1, \quad c = \frac{5}{2},$$

and any D one unit above the printed D -threshold. The printed constant threshold is positive; let d be half of it. At $x = -1$, all spatial derivatives of $W = d$ vanish, while the frozen operator equals

$$\begin{aligned} \mathcal{A}(d; u)(-1) &= -\chi d(V_0 - u(-1)) + d(1 - d) \\ &= -2d + d - d^2 \\ &= -d(1 + d) < 0. \end{aligned}$$

A subsolution requires $\mathcal{A}(d; u) \geq 0$. Hence the literal universal statement is false. The problem is not the nonsmooth interface of a minimum: the counterexample uses the separately stated constant subsolution and tests it at a regular point.

Repair. The missing content is a pointwise or uniform control of the frozen elliptic forcing. A transparent sufficient condition on a trap bounded by M is

$$|\chi| d^{m-1} M^\gamma \leq 1 - d^\alpha.$$

For example, the formalization proves the constant branch from

$$|\chi| M^\gamma \leq \frac{1}{2}, \quad 0 < d \leq \frac{1}{2}.$$

It also proves the $\chi = 0$ lower- and constant-subsolution slices under the corresponding D -scale hypotheses. The refutation is `not_Lemma_4_2` in `Paper1/Statements.lean`; the strengthened lemmas in the same file include `constant_subsolution_frozenWaveOperator_nonneg_of_half_bound_trap` and `Lemma_4_2_chi_zero_subsolutions_of_D_ge_one`.

3 Part I: boundedness and global existence

3.1 E3: Theorem 1.2, missing reaction guard

Printed range. The theorem assumes $a, b \geq 0$ and $\beta \geq 1$, which includes $a > 0, b = 0$, and asserts uniform boundedness in both its $0 < m < 1$ and small-sensitivity $m = 1$ branches.

Counterexample. On any smooth bounded Neumann domain, let $c_0 > 0$ and set

$$u(t, x) = c_0 e^{at}, \quad v(t, x) = \frac{\nu}{\mu} u(t, x)^\gamma.$$

The functions are spatially constant, so diffusion, chemotactic divergence, and all boundary fluxes vanish. The elliptic equation holds, and the parabolic equation reduces to $u_t = au$. Thus this is a positive global classical solution for every m, β, χ_0 , but

$$\|u(t)\|_\infty = c_0 e^{at} \longrightarrow \infty.$$

Equivalently, every positive solution in this branch satisfies

$$\frac{d}{dt} \int_\Omega u = a \int_\Omega u.$$

The counterexample already satisfies the $m = 1, \beta = 1, \chi_0 = 0$ smallness condition.

Correction and certificate. Under the standing nonnegativity assumptions, the maximal natural correction is

$$a = 0 \quad \text{or} \quad b > 0,$$

which excludes exactly $a > 0, b = 0$. The more conservative split used verbatim in the paper's proof is $(a = b = 0) \vee (a > 0 \wedge b > 0)$. The Lean refutations are `not_Theorem_1_2_intervalDomain_of_a_pos_b_zero` and `not_Theorem_1_2_intervalDomain_when_a_pos_b_zero` in `Paper2/IntervalDomainTheorem12Refutation.lean`. The corrected headline is `Theorem_1_2_intervalDomain_unconditional`.

3.2 E4: Theorem 1.3(iv), exponent-domain gap

The printed alternative (iv) assumes

$$\beta \geq \frac{1}{2}, \quad \alpha = 2m + \gamma - 2.$$

Section 5.4 invokes Proposition 2.2 at

$$s(P) = \frac{P + \alpha}{\gamma},$$

but that proposition requires $s(P) > 1$. At the critical identity this is equivalent to

$$P > 2 - 2m.$$

The proof starts from

$$q_* = \max \left\{ 1, \frac{N\alpha}{2} \right\},$$

so its stated route needs the missing condition

$$q_* > 2 - 2m.$$

This condition is not implied by the printed hypotheses. For example,

$$N = 1, \quad m = \frac{1}{4}, \quad \gamma = \frac{7}{2}, \quad \alpha = 2, \quad \beta = 1$$

satisfies $\alpha = 2m + \gamma - 2$, while

$$q_* = 1, \quad s(q_*) = \frac{6}{7} < 1.$$

Moreover $(N\alpha - 2)_+ = 0$, so the first smallness disjunct in (1.25) is automatic. The printed argument therefore invokes Proposition 2.2 outside its domain. This is a proof gap, not a counterexample to boundedness.

The proof-supported amendment is

$$\max\{1, N\alpha/2\} > 2 - 2m.$$

The formalization records the corrected paper-constant route in `Theorem_1_3_intervalDomainM_caseIV_with_paperConstants`. It also closes the printed theorem's actual one-dimensional conclusion for $m < 1$ by a separate guard analysis: the global conjunct is printed only for $m \geq 1$, while the bounded local branch is proved by `Theorem_1_3_intervalDomainM_of_m_lt_one`. The combined capstone is `Theorem_1_3_intervalDomainM_caseIV_all_m`. This alternative closure does not make the displayed Section 5.4 invocation valid; it explains why the repository can prove the final disjunct without silently endorsing that step.

3.3 E5: Proposition 1.1, maximal-continuation wording

The finite endpoint alternative must be attached to the distinguished maximal continuation. Read instead as a dichotomy for every positive finite local horizon T , it is false: for $a, b > 0$, the positive logistic equilibrium

$$u(t, x) = \left(\frac{a}{b}\right)^{1/\alpha}$$

is a classical solution on every finite interval, yet it neither blows up nor approaches zero there. The corrected statement constructs the supremum of reachable horizons and asserts the blow-up/floor-loss alternative only when that maximal horizon is finite; otherwise it glues the compatible local solutions into a global branch.

This is a wording and carrier clarification. The Lean witness against the over-literal producer is `not_legacyFiniteHorizonAlternativeProducer_of_positive_equilibrium`; the corrected endpoint-tail statement is `correctedProposition_1_1_intervalDomainM` in `Paper2/IntervalDomainMMaximalContinuationAlternative.lean`.

3.4 E6: Lemma 2.7, missing control at the initial endpoint

The literal statement assumes a differential inequality only for $t \in (0, T_{\max})$, but concludes a supremum bound on $[0, T_{\max})$. Take $T_{\max} = 1$ and a spatially constant function

$$u(t, x) = t^{-1}, \quad t \in (0, 1).$$

With $p = 2$, its mass quantity is $Y(t) = t^{-2}$, and

$$Y'(t) = -2t^{-3} \leq -t^{-3}.$$

Thus the interior differential inequality is satisfied for a suitable choice of the lemma's nonnegative constants (the formal witness uses $C_1 = C_2 = C_3 = 0$, $C_4 = 1$, and unit auxiliary exponents), while

$$\sup_{0 < t < 1} Y(t) = \infty.$$

No interior differential inequality can control an omitted initial trace.

The corrected form assumes continuity, or merely a finite bound, for the relevant L^p -mass at $t = 0$, then applies the scalar comparison on the closed interval. The exact certificates are `not_Lemma_2_7_intervalDomain` and `Lemma_2_7_intervalDomain_from_continuous_bound` in `Paper2/IntervalDomainLemma27Unconditional.lean`. In the PDE application the classical initial trace supplies precisely this missing endpoint information.

4 Part II: persistence and stabilization

4.1 E7: Theorem 2.1(1), omitted pure-decay branch

The printed part (1) assumes only $m \geq 1$ and claims a positive asymptotic spatial lower bound for every global bounded positive solution. The proof, however, treats $a = b = 0$ and $a > 0, b > 0$, omitting the allowed regime $a = 0 < b$.

For $c_0 > 0$, define

$$u(t, x) = (c_0^{-\alpha} + \alpha b t)^{-1/\alpha}, \quad v(t, x) = \frac{\nu}{\mu} u(t, x)^\gamma.$$

Again all spatial derivatives vanish, and the parabolic equation becomes

$$u_t = -b u^{1+\alpha}.$$

The orbit is global, bounded, and strictly positive for every finite time, but

$$\lim_{t \rightarrow \infty} \inf_{x \in \Omega} u(t, x) = 0.$$

Thus the printed part (1) is false in the pure-decay regime.

The proof-faithful guard is

$$(a = b = 0) \quad \text{or} \quad (a > 0 \wedge b > 0).$$

When $a > 0, b = 0$, a globally bounded positive solution cannot exist because its mass grows like e^{at} , so the implication is vacuous. The explicit Lean refutation is `not_Theorem_2_1_part1_intervalDomain_pureDecay`; the corrected statement is `Theorem_2_1_part1_corrected`, assembled into `Theorem_2_1_corrected_intervalDomainM`.

4.2 E8: Theorems 2.2–2.5, all-time C^1 decay

Equation (2.12) asserts a C^1 exponential estimate for every $t \geq 0$. Theorem 2.2 assumes only $u_0 \in C(\overline{\Omega})$, positive and close in L^∞ . Theorems 2.3–2.5 quantify over global bounded positive solutions and reuse (2.12), again with no uniform C^1 control at the initial time.

On $\Omega = (0, 1)$ with Neumann boundary conditions, take

$$u_{0,N}(x) = u_* + N^{-1/2} \cos(N\pi x).$$

For large N the datum is positive, has the exact minimal-model mass, and

$$\|u_{0,N} - u_*\|_{L^\infty} = N^{-1/2} \longrightarrow 0, \quad \|\partial_x u_{0,N}\|_{L^\infty} = \pi N^{1/2} \longrightarrow \infty.$$

Therefore an L^∞ -ball cannot have a uniform C^1 bound at $t = 0$. For merely continuous u_0 , the C^1 norm at $t = 0$ need not even be defined. Analytic-semigroup smoothing naturally has a $t^{-1/2}$ -type singularity as $t \downarrow 0$.

The correct statement separates two regimes:

1. if the initial perturbation is small in a fractional-power space $X^\theta \hookrightarrow C^1$, then the sectorial stability theorem gives an all-time strong-norm estimate;
2. from L^∞ -small or merely continuous data, one first evolves to some $t_0 > 0$, and then obtains C^1 exponential convergence for $t \geq t_0$.

The linear stability/instability threshold itself is unaffected.

The interval-domain obstruction is `not_intervalDomain_Theorem_2_5_original_allTime` in `Paper3/IntervalDomainSectorialCorrectedObstruction.lean`. The public headline theorems in the repository use `EventualExponentialC1ConvergenceWith` and orbit-dependent positive entry times. The same correction applies wherever Theorems 2.3–2.5 invoke (2.12).

5 Findings deliberately excluded from the paper errata

Formal proof assistants make over-quantification easy to detect. The following refutations are valuable interface audits, but they must not be reported as counterexamples to the source papers.

1. **Traveling-wave Lemma 4.1.** The Lean proposition `Lemma_4_1` in `Paper1/Statements.lean` omitted the printed negative-branch speed threshold and upper bound on M , and combined the two separately stated barriers into a global minimum. Consequently `not_Lemma_4_1` refutes that stronger Lean proposition. Likewise, `not_differentiableAt_upperBarrier_of_interface` observes that the minimum is not C^1 at its interface, whereas the paper states $e^{-\kappa x}$ only on one side and M separately. Neither theorem is a paper-level refutation.
2. **Traveling-wave Lemma 2.1.** `not_forall_Lemma_2_1` chooses fabricated norms and semigroup data in an abstract structure. The paper concerns the concrete heat semigroup, so this is a test of an overly universal API.
3. **Part I Theorem 1.3 FrontierData.** `not_IntervalDomainTheorem13FrontierData_of_real_regime` refutes an earlier structure containing arbitrary total functions and a global extension field. Those fields are not printed hypotheses. The genuine paper issue is only the exponent-domain gap E4.
4. **Part I Lemma 2.6.** An early abstract bootstrap interface stored a gradient frontier strong enough to encode the desired endpoint and did not expose the initial L^p -mass continuity. The repository proves `Lemma_2_6_intervalDomain_from_continuous_lp_bounds`, but it does not claim a mathematical counterexample to the PDE lemma as used in the paper.

5. **Positive-time mass and zero-time signal storage.** Earlier Lean trajectory records constrained the solution only for $t > 0$ while allowing the stored slices $u(0)$ or $v(0)$ to be changed. Refutations obtained by re-anchoring those slices diagnose the data model. The paper starts from a genuine initial datum and trace. The corrected Lean predicates use conserved positive-time mass or an explicit initial trace.

6 Reproducibility and interpretation

The calculations above were checked against the archived paper text, not against paraphrased Lean comments. The repository’s root build checks all headline imports. The intended interpretation of a Lean theorem name is:

- a theorem of the form `not_X` is evidence against the paper only after the hypotheses of `X` have been compared with the printed hypotheses;
- a proof of a corrected headline establishes the corrected mathematical statement, but does not retroactively validate a defective displayed step;
- an axiom-clean proof certifies deduction from the formal statement; it does not by itself certify that the formal statement has the intended semantics.

This distinction is why E2, E3, E6, and E7 are presented with explicit mathematical counterexamples, E1 and E4 as proof gaps, E5 and E8 as statement repairs, and the five interface findings in a separate section.